

IAC-25,A3,IP,1,x98063

Lunar Vitalis: Unlocking Resources for Sustained Exploration

**Gagana Yuvaraju^{a*}, Carlos Morales^b, Giuseppe Puleo^c, Sooraj Suresh^d, Saumya Shekhar^e, Andrea Bello^f,
Varsha Santhosh^g, Saatvik Sumanta^h, Ambrose Osadugbaⁱ, M Abdullah Khan^j, Maedeh Taherkhany^k, Eden
Abeselom Habteslasie^l**

^a Department of Aerospace Engineering, RV College of Engineering, Bengaluru, India; gaganay86@gmail.com

^b Space Generation Advisory Council (SGAC), Colombia

^c University of Naples "Federico II", Italy

^d James Watt School of Engineering, University of Glasgow, Glasgow, United Kingdom

^e TU Darmstadt, Germany

^f Sapienza University of Rome, Italy

^g Space Generation Advisory Council (SGAC), India

^h Space Generation Advisory Council (SGAC), India

ⁱ Space Generation Advisory Council (SGAC), Nigeria

^j Space Generation Advisory Council (SGAC), Bangladesh

^k Khajeh Nasir University of Technology, Iran

^l Ethiopian Space Science and Geospatial Institute, Ethiopia

* Corresponding Author

Abstract

Humans are closer than ever to establishing a permanent presence on the Moon, with upcoming missions like NASA's Artemis program paving the way for sustainable lunar exploration. The lunar south pole has drawn significant global interest, with its permanently shadowed regions (PSRs) believed to contain substantial water ice deposits. Water is a vital resource for supporting human life, producing oxygen, and serving as a key component for fuel generation, reducing dependence on Earth.

Recent studies suggest that two secondary craters within the Shoemaker Crater, located in the lunar south pole, hold the greatest potential for water ice deposits within the top meter of the surface. These PSRs exhibit both high circular polarization ratio (CPR) values, suggesting potential ice deposits, and high weight-equivalent hydrogen (WEH) concentrations, indicating the presence of hydrogen, likely in the form of water ice. These characteristics make them ideal sites for future sampling missions.

To explore these regions, this paper proposes a hopper/lander mission 'Lunar Vitalis' designed to drill into the lunar regolith, analyze its composition using spectrometers, and utilize microwave heating to extract water. Microwave heating is an efficient method for extracting subsurface ice, particularly in the Moon's vacuum environment, where conventional heat transfer is slow and the regolith acts as a strong thermal insulator. A microwave source will be placed inside a drilled hole, where it will sublimate subsurface ice into water vapor, which will then be collected, condensed, and stored in a cold trap system, eliminating the need for large-scale excavation.

This mission will provide essential ground-truth data on the lunar regolith's composition and water abundance in the Shoemaker Crater, validating orbital observations and advancing in-situ resource utilization (ISRU) technologies. By demonstrating an efficient and scalable water extraction technique, it will play a key role in enabling long-term lunar habitation and supporting future deep-space missions. The insights gained will not only inform upcoming lunar exploration efforts but also serve as a foundation for sustaining human presence on the Moon and beyond.

Keywords: In-Situ Resource Utilization, Lunar South Pole, Lunar Settlements, Permanently Shadowed Regions, Water Ice Extraction

Acronyms/Abbreviations

ΔV : Delta-V (change in velocity)

g: Gravitational acceleration

m₀: Initial wet mass

H: Apex height

D: Horizontal distance

Isp: Specific Impulse

N: Newton

T/W: Thrust-to-weight ratio

RCS: Reaction Control System

CBE: Current Best Estimate

MEV: Maximum Expected Value

CONOPS: Concept of Operations

VTVL: Vertical Takeoff and Vertical Landing

LOX/LH₂: Liquid Oxygen/Liquid Hydrogen

LOX/CH₄: Liquid Oxygen/Liquid Methane (also referred to as methalox)

MMH/MON-25: Monomethylhydrazine/Nitrogen
Tetroxide with additives
NASA: National Aeronautics and Space Administration
JPL: Jet Propulsion Laboratory
CORE: Common Repository
MIT: Massachusetts Institute of Technology
ISE: Integrated Service Engine
TVC: Thrust Vector Control
GN&C: Guidance, Navigation, and Control
FOD: Foreign Object Damage
SCAPA: Self-Contained Atmospheric Protective
Ensemble

1. Introduction

1.1 Background and Motivation

The prospect of sustained human presence on the Moon has placed a renewed focus on the identification and utilization of local resources. Water ice, in particular, is indispensable, not only as a life-supporting resource for drinking and oxygen production but also as a key input for propellant production, drastically reducing the need for Earth-based resupply. This makes the confirmation and utilization of lunar volatiles a central driver for upcoming exploration initiatives.

Current and planned missions, such as NASA's Artemis program and international lunar exploration efforts, are increasingly directed toward the Moon's polar regions. These missions aim to validate the presence of volatiles, demonstrate in-situ resource utilization (ISRU) technologies, and establish the infrastructure required for long-term habitation and deep-space missions.

Three-dimensional mapping of the lunar polar regions ($\geq 70^\circ$ latitude), using integrated radar, optical, and neutron data from NASA's Lunar Reconnaissance Orbiter (LRO), has shown that subsurface ice ($\sim 1\text{--}3$ m depth) covers 5–8 times more area than surface ice (≤ 1 m depth) at both poles. The total areas of water ice-bearing regions in the northern and southern polar regions are $\sim 1100 \pm 74$ km² and $\sim 562 \pm 54$ km². The key parameters considered to delineate regions with the utmost likelihood of surface (up to ~ 1 m depth) and subsurface ($\sim 1\text{--}3$ m depth) water ice occurrence are: (1) anomalous craters with a high Circular Polarization Ratio (CPR) derived from SAR observations, (2) PSRs, and (3) areas with a high Water Equivalent Hydrogen (WEH) content, as derived from epithermal neutron measurements [1].

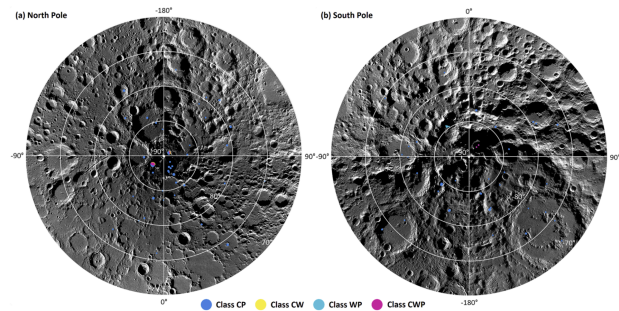


Fig. 1. The polar mosaic derived from the LRO Wide Angle Camera (WAC) of lunar (a) North polar region (70°N onwards) and (b) South polar region (70°S onwards), showing the locations of water ice regions belonging to various Classes. Regions within Blue, Yellow, Cyan, and Pink colored polygons belong to classes CP, CW, WP, and CWP, respectively. The images are in lunar polar stereographic projection [1].

The CWP regions (high CPR anomalous craters with high WEH concentrations within PSRs) in the lunar south pole are considered to be the most promising sites as they represent the highest probability of hosting stable, near-surface to subsurface water ice, making them priority targets for future exploration and in-situ resource utilization. This research explores a mission to these sites, aiming to confirm the presence of near-surface and subsurface water ice, demonstrate microwave-assisted extraction and cold-trap storage, and validate technologies for mobility and thermal resilience in extreme polar conditions.

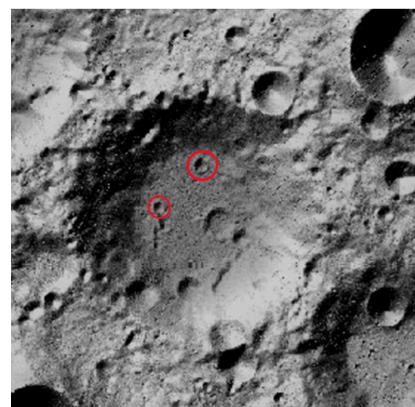


Fig. 2. CWP regions in the Shoemaker crater.

1.2 Scientific Objectives

Lunar Vitalis is designed to explore two secondary craters within Shoemaker Crater, targeting regions with the highest potential for near-surface water ice. The mission combines drilling, spectroscopic analysis, and microwave heating to confirm the presence and form of ice, extract it efficiently, and capture it in a dedicated

cold-trap system. By validating orbital data with ground-truth measurements and demonstrating a scalable extraction technique, Lunar Vitalis advances ISRU technologies essential for future exploration. The hopper-based architecture further enables mobility across PSRs, expanding sampling coverage while testing survival strategies in the extreme thermal environment of the lunar south pole. Together, these objectives address the dual challenge of resource prospecting and technology demonstration, paving the way for sustainable human habitation and deep-space missions beyond the Moon.

2. Mission Objectives and Requirements

2.1 Mission Objectives

Table 1. Primary Mission Objectives

ID	Description
PMO 01	To validate the presence, form and quantity of water ice within the mission site.
PMO 02	To study and operate the mission in two secondary craters of Shoemaker Crater.
PMO 03	To demonstrate an excavation technique based on a drill mechanism combined with microwave heating.
PMO 04	To demonstrate the capability to store water in cold trap systems.
PMO 05	To prospect and validate the properties of the lunar regolith within the mission site.
PMO 06	To demonstrate the hopping capability of the lunar rover to move within the mission site.
PMO 07	To design and validate a system architecture capable of surviving X Earth Days in the lunar environment.
PMO 08	To maximize the mass-by-cost ratio and the scientific return of the mission.

Table 2. Secondary Mission Objectives

ID	Description
SMO 01	To validate the presence of hydrated minerals within the mission site.
SMO 02	To evaluate and assess the in-site water content concentration, and compare it with orbital data.
SMO 03	To validate outdated databases and methods about water presence in lunar south pole.
SMO 04	To validate lunar ISRU systems for future sampling missions.
SMO 05	To strengthen international space cooperation between space agencies and private companies.

The primary and secondary mission objectives are listed in Table 1 and Table 2 based on the scientific objective the mission aims to achieve.

2.2 Mission Requirements

Based on the mission objectives, the mission requirements are listed in Table 3.

Table 3. Mission Requirements

Req ID	Title	Description
LV-MIS-0010	Water Ice Prospection	The Mission shall investigate and assess the presence, the form and the quantity of water ice in specific regions of the lunar south pole.
LV-MIS-0020	Water Ice Extraction	The Mission shall demonstrate the technological capabilities of drilling excavation techniques and microwave heating to extract water ice from the lunar soil.
LV-MIS-0030	Water Ice Storage	The Mission shall demonstrate the technological capabilities of storing the extracted water in cold trap systems.
LV-MIS-0040	Region of Interest	The Mission shall be carried out within the Shoemaker Crater; a permanent shadow region located in the lunar south pole.
LV-MIS-0050	ISRU Robotic System	The Mission shall make use of a single robotic system able to fit a commercial lunar lander and perform the required ISRU activities.
LV-MIS-0060	Rover Hopping Capabilities	The Mission shall make use of a hopping robotic system able to move and operate within the mission site.
LV-MIS-0070	Environmental Threats	The Mission shall be design to survive both interplanetary transfer and lunar environment.
LV-MIS-0080	Lunar Soil Prospection	The Mission shall validate the properties of the lunar soil by means of analysis and prospection.
LV-MIS-0090	Design Philosophy	The Mission shall maximize its mass-by-cost ratio, making use of COTS components and consolidated launch/landing strategies.

2.3 System Requirements

Req ID	Short Title	Requirement Description	Backward Links
LV-SYS-0010	Excavation Data Source	The System shall rely on databases and resource maps to determine the excavation sites for water ice and other in-situ resources.	LV-MIS-0010 LV-MIS-0080
LV-SYS-0020	SWIR Spectrometer	The System shall make use of a Short-Wave Infrared (SWIR) Point Spectrometer to confirm the presence of condensed water ice within the collected sample.	LV-MIS-0010
LV-SYS-0030	Rover Cameras	The System shall make use of a multiple cameras to map crater interior, verify drilling site and monitor vapor transport up to the storage unit.	LV-MIS-0010
LV-SYS-0040	Warm Optics Box	The System shall adopt illumination and packaging practices to structure a warm payload optics box.	LV-MIS-0010
LV-SYS-0050	Drill Instrument	The System shall make use of a lunar regolith drill to access the water ice site and prevent volatile loss through RF borehole shutter.	LV-MIS-0020
LV-SYS-0060	Maximum Extraction Depth	The System shall extract condensed water ice reaching a maximum depth of 1 m below the surface regolith.	LV-MIS-0020
LV-SYS-0070	Microwave Heating	The System shall extract condensed water ice using an integrated microwave applicator, capable of sublimating ice-bearing regolith or pure ice into water vapor.	LV-MIS-0020
LV-SYS-0080	Cold Trap	The System shall transport the extracted water in a thermally condenser-first cold trap via heated vapour line.	LV-MIS-0030
LV-SYS-0090	Vapor Transport	The System shall capture and condense the transported sublimated water vapor into a storage unit, by achieving a stable frost growth under low-pressure conditions.	LV-MIS-0030
LV-SYS-0100	Mission Duration	The System shall be able to perform its mission operations and activities for a total time of 2-4 Earth days in lunar environment.	LV-MIS-0040
LV-SYS-0110	Identified Mission Sites	The System shall be able to inspect and operate in at least two sub-craters or sub-regions within the Mission Site.	LV-MIS-0040
LV-SYS-0130	Lander Compliance	The System shall be compliant with the constraints imposed from the commercial Novac-Lander.	LV-MIS-0050 LV-MIS-0090
LV-SYS-0140	Rover Mass	The System shall include a single rover, whose total mass shall be lower than or equal to 500 kg, including design margins.	LV-MIS-0050 LV-MIS-0090
LV-SYS-0150	Rover Volume	The System shall include a single rover, whose overall envelope shall be lower than or equal to 3 m ³ , including design margins, in stowed configuration.	LV-MIS-0050
LV-SYS-0160	Payload Housing	The System shall host the Payload on board the rover, ensuring its protection from the lunar environment, its power supply during stand-by and nominal modes, and the management of commands and scientific outputs, throughout its operational life.	LV-MIS-0010 LV-MIS-0020 LV-MIS-0030 LV-MIS-0050
LV-SYS-0170	Rover EPS Function	The System shall include an Electrical Power Subsystem (EPS) on board the rover, which shall be able to generate, store, condition and distribute the electrical power used by the rover in all operational modes.	LV-MIS-0050 LV-MIS-0090
LV-SYS-0180	Rover TCS Function	The System shall include a Thermal Control Subsystem (TCS) on board the rover, which shall ensure all bus units and payload components remain within their allowable temperature and radiative ranges.	LV-MIS-0050 LV-MIS-0070 LV-MIS-0090
LV-SYS-0190	Rover CDDHS Function	The System shall include a Communications, Command and Data Handling Subsystem (CDDHS) on board the rover, which shall regulate telecommand reception and telemetry transmission, acquire and store platform and payload data, and perform autonomous supervision and management tasks.	LV-MIS-0050
LV-SYS-0200	Rover Hopping Function	The System shall include a Ballistic Hopping Propulsion Subsystem on board the rover, which shall allow ballistic hopping between the mission site, attitude and trajectory control, and autonomous navigation with corresponding hazard avoidance.	LV-MIS-0050 LV-MIS-0060 LV-MIS-0090
LV-SYS-0210	Hopping Capability	The System shall be able to cover at least 0.5 km in each hopping actions.	LV-MIS-0060

Fig. 3. System requirements with backward links

3. Mission Operations

The Lunar Vitalis mission proceeds through three major operational phases: initialization, science and mobility operations, and closeout; spanning a total of 2-4 Earth days within the two subcraters in the Shoemaker Crater. Operations are highly autonomous due to Earth-Moon communication delays and periodic blackout windows when direct contact is obstructed.

Phase 1 (Initialization): Following delivery by the host lander, the hopper separates in a stowed configuration and deploys its systems for readiness checks. Subsystems for power, propulsion, thermal control, and guidance are validated in the extreme cryogenic PSR environment. Instruments including the TRIDENT-class drill, microwave heater, cold trap, and spectrometer are deployed and calibrated. A spring-launched router is ejected toward the crater rim to establish a communication relay. A shallow 10 cm drill test, accompanied by a brief microwave heating run and spectral check, verifies functionality. This phase establishes initial communications, validates positioning with Terrain Relative Navigation (TRN) against preloaded maps, and ensures all subsystems are operational before science activities commence.

Phase 2 (Science Operations and Hopping): At the initial site and the subsequent hopped site, the hopper performs its core scientific campaign. Each site is accessed through a vertical takeoff and landing sequence with ballistic arcs of 2-6 km, guided by LIDAR, terrain-relative navigation, and RCS stabilization. Upon soft touchdown, drilling proceeds in incremental 10 cm bites to depths of up to 1 m. The integrated microwave applicator then applies

150W-250W bursts to extract volatiles, which are sealed and directed into a cold trap for capture and analysis. Spectral measurements in the 1.0-2.1 μm range confirm the presence of water ice, with per-bite yields logged along with thermal and environmental data. Each site typically requires 12-24 hours to complete sampling and analysis. Across the mission, a total of two to six sites are visited, requiring five hops and yielding an estimated dataset of 10-20 GB including spectra, images, and environmental logs.

Phase 3 (Data Relay and Closeout): At the final site, the hopper performs an additional sampling run before preparing for mission closeout. All stored data are transmitted via the deployed router to the Deep Space Network, with priority given to critical ice confirmation results. If resources allow, an optional final hop to a sunlit crater rim may be conducted to extend operations. The mission concludes with the safe shutdown of instruments and transition into a quiescent state, leaving a complete dataset that validates water ice abundance, extraction efficiency, and ISRU feasibility.

The Lunar Vitalis hopper completes two to six sampling sites in 2-4 or 7-12 Earth days, consuming ~300 kg of propellant across all hops, and returning ~20 GB of scientific data. Success criteria include confirmation of water ice at multiple sites, demonstration of extraction efficiencies exceeding 67%, and delivery of comprehensive environmental and operational data to Earth.

4. Scientific Payload

4.1 Science Goals

The mission aims to achieve the following science goals:

1. Confirm H₂O as condensed ice in regolith to ~1 m by extracting vapour and spectrally identifying condensed ice on a controlled condenser (not bound OH or gas-phase signatures) [2-5].
2. Validate orbital interpretations by providing ground-truth confirmation of ice (not OH) to anchor M³/LCROSS/Trailblazer-style interpretations [6].
3. Quantify extraction performance: per-bite and cumulative yield, power/dwell dependence, energy cost (Wh g⁻¹), and condenser P/T set-points that sustain desublimation as frost grows [7-10].
4. Characterise vertical variability and context: map shallow distribution with depth via repeated bite cycles; record thermal/regolith context relevant to volatile retention [2,3].

4.2 Payload Architecture

The payload architecture is designed around an integrated drill-microwave system capable of reaching

depths of up to 1 m in incremental bites, with the RF radiator positioned at the drill bit and a hollow auger serving as the internal vapor channel [5]. A borehole packer and shutter at the collar seal the annulus, directing vapor into the auger channel and subsequently through a heated duct, thereby preventing venting into the vacuum. The vapor path, insulated and equipped with Kapton heaters, ensures that the water vapor remains above its dew point until it reaches the cold trap, eliminating the need for a carrier gas [2, 3]. The sub-zero cold trap provides a dedicated condenser as the unique cold sink, enabling controlled condensation at a location observable by the optical system [2]. A VNIR/SWIR spectrometer, positioned to view the condenser plate, confirms the presence of H₂O ice, with no gas-phase analysis included in the baseline design [4].

4.2.1 Drill System Specifications (with Integrated RF Head)

The TRIDENT-class rotary-percussive drill, derived from VIPER heritage [5], is a proven 1m-class bite-drilling system capable of both rotary and percussive operations. The drill as in Fig. 3. features a 25.4 mm bit diameter, maximum depth of 1020 mm, spin rate of 120 RPM, and an average torque of 5.5 Nm, with percussion delivered at 972 blows per minute at 2 J per blow. The typical auger power is 87 W, and it operates with a feed stroke of 1240 mm and a deploy stroke of 380 mm, with a total mass of approximately 29 kg (22 kg drill with locks plus 7 kg avionics). The integrated concept, similar to ProSEED [11], includes a microwave radiator at the drill tip, with the hollow auger serving as the vapor channel (ID $\geq$ 10–12 mm). A packer/shutter at the collar seals the annulus to prevent volatile loss, while thermal standoffs and RF chokes protect the bearings and avionics. The TRIDENT drill has been extensively tested in flight-like configurations, including TVAC and cryogenic endurance campaigns of up to ~24 hours with multiple holes, establishing it as a low-risk solution for subsurface access. For the current mission, only two boreholes are required, with a budget of up to ~10 hours per 1-m hole in worst-case cohesive or icy soils, while operations in more benign materials are expected to be faster.

The drill tip stack integrates mechanical and RF functionalities in a compact configuration. It consists of the drill bit, a thin dielectric RF window (sapphire or fused silica, thickness ~1–2 mm), and a ring-slotted or end-fire microwave radiator, with the hollow auger serving as the vapor channel. The coaxial feed includes RF chokes, and a packer/shutter at the collar ensures annulus sealing. The structure combines a Cu/Ag-plated RF cavity with a Ti or stainless-steel mechanical frame, while low-loss dielectrics are used for critical RF components.

4.2.2 Microwave Heating and Cold Trap

The system heats at 2.45 GHz using a 150-250 W GaN amplifier chain that feeds a ring-slotted or end-fire radiator positioned directly behind the drill bit, protected by a thin sapphire or fused-silica RF window. Heating is applied in 10-20 minute bursts with slow axial dithering of the probe to sweep the hot zone and prevent local dry-out, producing just enough vapor for plate capture. This power and timing regime is supported by laboratory results at the same frequency: integrated rigs achieved approximately 67% capture in ~25 minutes (~90% by ~35 minutes), while campaigns with arrangement-controlled conditions (permafrost vs. granular soils) delivered 83-97% capture in 60 minutes with dynamic-mass tracking [8]. Control is maintained via line and plate thermometry, with forward/reflected power interlocks and packer-side pressure feedback, minimizing time and energy while remaining within the 150–250W envelope [2,3,7,8]. Near-quantitative capture to a cryogenic trap has been demonstrated in LN₂ “Moon-in-a-Bottle” experiments [9,12].

Vapor is maintained above 0 °C until it reaches the collection plate using Kapton heaters and insulation, with line temperatures held between +20 and +40 °C. On the Moon, the near-vacuum environment allows the same hot-to-cold gradient to drive capture, while ground or analog testing can employ a small valve or pump to bias the line at ~100–500 Pa, reproducing low-pressure frosting regimes used in models [2,3].

The cold trap is a compact, sealed enclosure (200 × 200 × 200 mm) containing a high-conductivity condenser plate (200 × 200 × 100 mm) made of Cu or Al with a protective Au/Ni finish, mounted on low-conductance standoffs to act as the sole cold sink. A sapphire or fused-silica window with a thin heater is maintained at 0-40 °C to prevent frosting and maintain optical clarity, while interior walls and baffles are low-emissivity and actively warmed so vapor nucleation occurs only on the condenser. The vapor line enters tangentially with a short, smooth path, and all joints are gasketed to preserve pressure near the plate.

Warm transport keeps the vapor line at +20 to +40 °C, while the condenser is held at –80 to –100 °C (qualified to –173 °C) to achieve near-unity sticking, negligible re-sublimation, and stable frost growth under low-pressure conditions. The plate experiences radiative load from the warm window and walls, conductive load through mounts, and latent load from freezing and sublimation. Heat is rejected via a dedicated miniature cryocooler or a space-view radiator, with thin-film heaters trimming walls and the window to prevent parasitic frosting. Sensors include RTDs on the plate, a thermistor on the window, thermocouples on the line, and an absolute pressure tap near the plate; an optional QCM monitors deposition

rate during tests. This architecture and set points align with low-pressure cold-plate experiments and ice-growth models, and capture efficiency is de-risked by LN₂ cold-trap demonstrations showing near-quantitative collection [2,3].

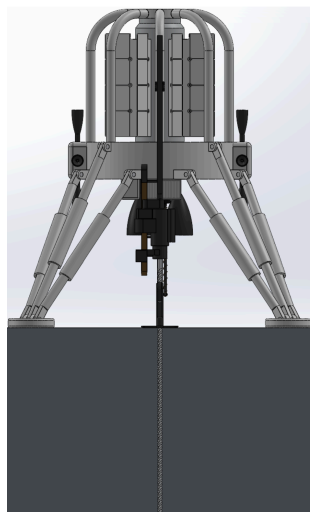


Fig. 4. Drill System

4.2.3 Spectrometer and optics

A compact, sealed, lamp-lit reflectance node is used to confirm H₂O as ice on the condenser, employing a single SWIR point spectrometer (1.0-2.1 μm) and a miniature VIS context camera. The system targets the 1.25, 1.5, and 2.0 μm absorption features, which strengthen rapidly even at very low ice abundances ($\sim 0\text{-}5\text{ wt}\%$), enabling a simple “ice present” determination at modest resolution and SNR. Illumination and geometry follow proven practice, using a mini NIR lamp or NIR LEDs at a fixed phase angle of $\sim 45\text{-}60^\circ$, ensuring the 1.5 and 2.0 μm bands remain robust against variations in grain size and texture [2,3]. The subsystem detects condensed ice rather than gas-phase volatiles or liquid water, as liquid is unstable under lunar vacuum and gas analysis would require additional valves or mass spectrometry, while condensed-phase spectra in this wavelength range are strong and unambiguous [2,3]. Interface controls including warm vapor lines, heated window and baffles, and a single cold sink at the plate ensure that deposition is localized precisely where it is measured, consistent with low-pressure cold-plate experiments and vacuum ice-growth models [5].

The spectrometer design is guided by prior studies and performance targets. The spectral window must cover 1.0-2.1 μm (minimum) to capture the 1.25, 1.5, and 2.0 μm ice absorption features, with 2.5 μm as a goal to provide additional margin [9]. Resolution is required to be $\leq 20\text{ nm}$ FWHM to enable continuum-removed band-depth analysis of the broad

ice absorptions [6]. Sensitivity must allow detection of band depths as low as 2-3 % at minimal ice abundance, with a design target SNR ≥ 150 near 1.5 μm for reflectances of $\sim 10\text{-}20\%$ [4]. Illumination and geometry follow established practice [3,4], with a fixed phase angle of $\sim 45\text{-}60^\circ$ using lamp-lit reflectance (mini tungsten-halogen lamps or NIR LEDs) consistent with DOC heritage [4,8]. Thermal and contamination control sets the window and baffles at $0\text{-}40^\circ\text{C}$, the vapor transport line at $+20\text{-}40^\circ\text{C}$, and the condenser at sub-zero temperatures, ensuring the plate acts as the sole cold sink [2,3,9,10].

4.3 Payload Operations Concept

Operations begin with preheating and system checks: the condenser is set to -80 to -100°C , window and baffles are maintained at $0\text{-}40^\circ\text{C}$, and the transport line is heated to $+20\text{-}40^\circ\text{C}$. Optics are warmed up, instrument health checks are performed, and a dark frame is acquired. Access to the subsurface is achieved by drilling in $\sim 10\text{ cm}$ bites to depths of $0.5\text{-}1.0\text{ m}$, pausing as needed for thermal stabilization and cuttings removal. The packer/shutter is then deployed at the collar, and seal integrity is verified using a pressure tap. Microwave extraction is performed at 2.45 GHz with $150\text{-}250\text{ W}$ in $10\text{-}20$ minute bursts while slowly stroking the applicator a few centimeters to sweep the hot zone, logging forward/reflected RF power and tool/line temperatures. Vapor transport maintains the internal channel and external duct above 0°C (target $+20\text{-}40^\circ\text{C}$) without the use of fans in vacuum. Cold capture is achieved by holding the condenser at -50 to -80°C while keeping walls and window warm to prevent parasitic frosting; optionally, the plate shutter may be opened after $5\text{-}10\text{ s}$ to bypass initial dust. Spectral confirmation follows a “detect-and-stop” approach: dark and lamp-on spectra are acquired with a VIS context image, continuum-normalized, and band depths at 1.25, 1.5, and 2.0 μm are computed. If at least two of the three bands exceed 2-3 %, H₂O ice is confirmed, and RF power is stopped immediately; if below threshold, deposition may continue for 1-2 minutes or the plate may be annealed (e.g., from $\sim -80^\circ\text{C}$ to $\sim -50^\circ\text{C}$) before re-measurement. Data downlink begins with decision-epoch spectra, followed by housekeeping (plate/line temperatures, pressure, RF power) and 1-2 VIS context images, with full logs queued as bandwidth allows. Shutdown and retraction conclude operations, ramping RF to zero, securing the packer/shutter, retracting the tool, and returning heaters to standby set-points.

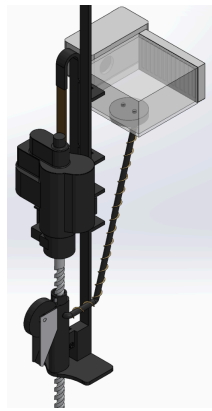


Fig. 5. Payload Set-up



Fig. 6. Microwave Heater tip

5. Spacecraft Design (Baseline Concept)

5.1 Structure and Mechanism

5.1.1 Design Rational

The Hopper module is designed to execute the mission while accommodating the constraints of its major subsystems. The electrical power system is based on an RTG modeled after the MMRTG [13]; its dimensions dictated the structural envelope, ensuring that the radiator fins have an unobstructed view of space for effective heat rejection. Scientific operations rely on the TRIDENT rotary-percussive drill [14], whose assembly dimensions further constrain the size of the main structure. Mobility is provided by two L3Harris-200 main engines and two RCS packs, each with four L3Harris-5 engines using MMH/MON-25 propellant [15]; the structure accommodates the necessary propellant tank volume based on the Hopper's total mass. Envelope constraints from the Nova-C lander define the stowed configuration, while communications and C&DH requirements specifically, a router placement near the crater rim led to a dedicated deployment mechanism on the lander. The Hopper has a total mass of approximately 500 kg, with stowed dimensions of $991 \times 1898 \times 1162$ mm and deployed dimensions of $1642 \times 1718 \times 1642$ mm.

5.1.2 Design Description

The Hopper's main structure incorporates a cylindrical volume with a radius of 400 mm and a height of 200 mm, which is sized to house both the propellant storage cylinder and the avionics. An arching

support formed from eight cylindrical members surrounds the RTG and is arranged to ensure the RTG radiator fins retain an unobstructed view to space for effective heat radiation.

Each hopper leg is a three-piece assembly consisting of a 75 mm diameter midsection and two 50 mm diameter end sections. This arrangement enables telescopic extension and retraction to adapt to varying terrain and improve shock absorption during touchdown. The leg attachment to the primary structure permits rotation so the legs may fold to a vertical orientation, minimizing stowed volume.

The footpads are 250 mm diameter cylindrical elements designed to provide stable ground contact during landing and hopping operations. Their geometry allows them to partially penetrate the regolith, increasing surface adherence and reducing the risk of slippage on uneven terrain.

The router deployment mechanism consists of three preloaded springs that, when released, generate sufficient impulse to place the router on a parabolic trajectory toward the crater rim. This passive launch approach was selected due to its mechanical simplicity, low mass, and high reliability compared to more complex actuation systems. The spring arrangement is designed to ensure stable ejection dynamics, preventing tumbling during flight and allowing the router to land in the desired location.

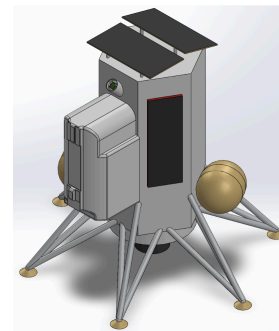


Fig. 7. Lander isometric view

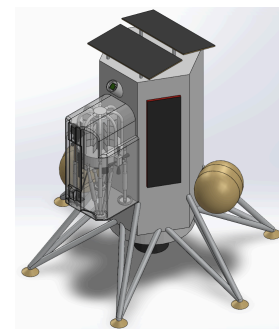


Fig. 8. Lander isometric view with Hopper inside

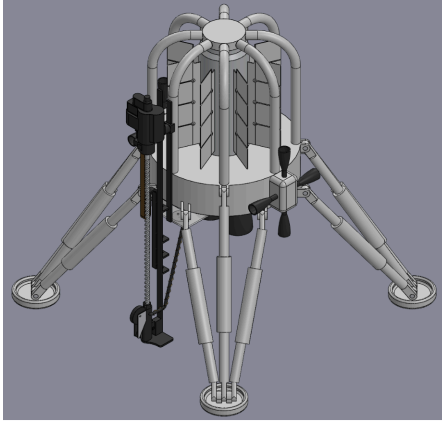


Fig. 9. Hopper isometric view

A router is deployed to maintain communication between the Hopper and Earth after the Hopper detaches from the lander. Deployment is achieved using a spring-based system, as illustrated in Fig. 6 and Fig. 7, which imparts a controlled impulse to place the router along a stable parabolic trajectory toward the crater rim. This passive spring mechanism was chosen for its mechanical simplicity, low mass, and high reliability, ensuring accurate placement.

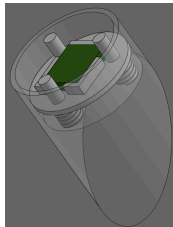


Fig. 10. Router Launcher compressed state

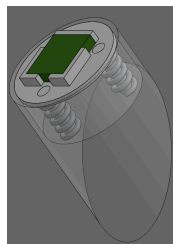


Fig. 11. Router Launcher uncompressed state

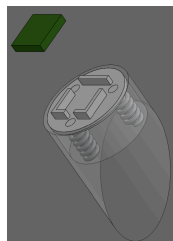


Fig. 12. Router Launched

5.2 Thermal Control System

5.2.1 Lunar Thermal Environment

Lunar craters in the southern pole, while generally cold enough to preserve water ice and other volatiles, are not thermally static. Surface temperatures in PSRs are as low as $\sim 25\text{K}$, with calibration uncertainties suggesting the possibility of values below 20K [16]. Some literature also indicates that PSRs in the lunar south pole usually have stable temperatures between $38\text{--}40\text{ K}$, with maximum values that are consistently below the water ice sublimation threshold ($\sim 110\text{ K}$).

A simplified model based on the available data found in the literature is used to depict the thermal environment of Shoemaker Crater. The assumed parameters for PSRs are compiled in Table 1, along with values for illuminated lunar terrain for comparison. Since surface reflectance contributes very little to the overall heat balance when there is no direct insolation, the effective albedo is assumed to be zero. Similarly, the scattered and refracted light from nearby illuminated terrain reaches the crater floor at intensities far below the threshold of thermal significance, setting the effective solar flux to zero. The planetary infrared flux, which is derived from re-radiation by nearby sunlit crater walls and topography, is the main contributor, albeit weak on the order of a few W/m^2 and in this study, it is conservatively estimated to be around 5.2 W/m^2 .

Parameter	Illuminated Lunar Terrain (Non-PSR)	Permanently Shadowed Region (Shoemaker Crater)	Sources
Lunar Surface Temperature (K)	95.15 – 390.15	Stable: $\sim 38\text{--}40$ minima: $\sim 20\text{--}25$; maxima: <110	[2] [4] [5] [6]
Solar Flux (W/m^2)	1414 (subsolar peak)	0	[1] [7]
Albedo	0.076 – 0.297	0	[7] [8]
Planetary IR Flux (W/m^2)	1314 (subsolar peak)	5.2	[7]
Imposed heat flow (W/m^2)	0.031	0.031	[9]
Deep Space Temperature (K)	$2.7\text{--}3\text{ K}$	$2.7\text{--}3\text{ K}$	CMB

Fig. 13. Thermal environment conditions on lunar surface for PSR vs non-PSR

5.2.2 Lunar Thermal Environment

The Thermal Control System (TCS) of Lunar Vitalis must guarantee the survival of critical subsystems in the extreme cryogenic environment of Shoemaker Crater while enabling operational and scientific objectives. The main requirements of TCS can then be derived as (i) ensuring the survival of all subsystems by maintaining them within operational thermal limits, (ii) facilitating effective and controlled microwave heating for the sublimation of water ice from the regolith, and (iii) preserving sufficiently low temperatures in the cold trap to condense the vapor. For this study, PSRs ambient temperature range is taken as $40\text{--}110\text{ K}$, which

is significantly below the survival limits of most components.

A crucial challenge lies in balancing contrast between the cold external environment and internal dissipation of the hopper system. Active phases such as drilling, microwave heating, or hopping generate substantial amounts of transient heat load and it is important to prevent this excess heat from propagating into nearby thermal zones of the hopper, where it could disrupt stability or compromise sensitive systems. Even during idle periods, mission-critical subsystems dissipate a modest but continuous heat flux, requiring stable regulation.

Most elements can be treated at the “box level,” due to having the same enclosure or having similar thermal limits but to avoid overestimating the thermal limits, outliers in subsystems must be considered separately. For instance, the external antenna is the only part of CC&DH that can work in a much wider range of conditions while the drill and microwave head must be dealt with separately as they generate a lot of heat and consume a lot of power. Similarly, cold trap is a separate functional element that is not included in the subsystem envelope and is treated separately in ISRU-specific design considerations as its requirements are different from those of standard electronics. In the Electrical Power System (EPS), RTG constitutes a self-heated outlier, but other components are considered within a box-level envelope. Propulsion hardware presents additional risks, as propellant freezing in PSR conditions could threaten reliability and may pose challenges to the valve integrity. Fig.8. summarizes the adopted operational and survival temperature ranges for each subsystem [17] [18] [19].

Subsystem		Typical Temperature Ranges (°C)			
		Operational		Survival	
		Min	Max	Min	Max
CC&DH	Box	-20	50	-40	60
	Antenna*	-100	40	-120	120
Science & Payload	Box	-20	40	-30	60
	Drill*	-55	+125	-120	70
	MW	-20	50	-20	70
Propulsion & Ballistic Hopping		0	40	-10	50
Electrical Power System		-10	40	-20	50

Fig. 14. Thermal Operating and Survival limits for subsystems

5.2.3 Thermal Design

At the system level, a zoned architecture methodology is adopted where subsystems with comparable thermal limits are grouped into a “box” while treating outlying elements with wider or narrower thermal limits separately. Thermal design is governed by several principles mainly related to subsystem survivability and is divided into three distinct zones:

warm, cold, and hot. Grouping these components together with similar operational ranges reduces the risk of power growth exceeding capacity. The warm zone encompasses subsystems with moderate operational and survival limits, such as CC&DH, EPS components, and other avionics. The hot zone includes components that dissipate heat, such as drill, microwave heating apparatus, and RTG. The cold zone consists of a cold trap and radiator that must always remain under the sublimation threshold of water ice (~110 K) to enable efficient volatile capture [20].

A prudent selection of TCS components is carried out for Lunar Vitalis, by examining a plethora of both passive and active thermal solutions. Most passive thermal solutions considered for the hopper operate by optimizing surface properties by reducing radiative heat losses through the application of low-emissive Multilayer Insulation (MLI) blankets (15-30 layered), specialized thermal films, and thermal coatings. Radioisotope Heater Units (RHU) could also be employed to further improve the effectiveness of passive solutions, as these heater units provide the heat energy of a decaying radioisotope without consuming power [21,22]. These passive solutions are implemented to enhance radiative isolation while making strategic use of RTG waste heat to offset heater power demand. Moreover, thermal isolation between warm, cold, and hot zones is achieved by compartmentalizing high dissipation components such as the drill and microwave heater from the warm avionics bay through using low conductivity mounts and dedicated heat straps. Annealed Pyrolytic Graphite (APG) and customized doublers and spreaders could be utilized to provide efficient transport of heat away from localized hot spots such as RTG without requiring active pumping or moving parts. In tandem with these passive thermal solutions, some active thermal control strategies are also adopted to ensure precise regulation of subsystem temperatures during various mission phases to ensure system resilience, ensure redundancy, and manage transient loads in PSR. Several electrically resistive heaters are utilized specifically; Kapton heaters, like the Tempco SHK series, are internally mounted on the enclosure with avionics with aluminum overtaping to prevent local heat spots and to maintain operation temperature for prolonged usage in the PSR environment. Externally, flexible heaters are wrapped around vapor transport lines to prevent condensation and re-freezing of water vapor en route to the cold trap. The drill utilized in this hopper is based on the TRIDENT drill, which comes with a 40W survival heater within its design to ensure its structural as well as operational integrity [17,23].

In addition to these components, thermal regulation is realized by an integrated network of thermostats and heat switches. The thermostats autonomously switch

the heaters on and off at predefined set points to maintain thermal margins. Heat switches passively modulate thermal conductance, enabling radiator coupling and decoupling [17,24] depending on the temperature. Accurate thermal monitoring is enabled by a suite of sensors composed of silicon diodes, thermistors, thermocouples, and platinum resistance thermometers (PRT). Silicon diodes are utilized on the coldest interfaces, such as cold trap and radiator, where they can operate with high efficiency in cryogenic regimes less than 60 K. Thermistors, specifically 10k negative temperature coefficient (NTC) thermistors and the Renesas ISL73590SEH [25], are suited for monitoring nominal temperatures in avionics bay. In areas such as drill-head where high temperature gradients are present with rapid fluctuations, a thermocouple could be feasible. Finally, PRTs such as the Collins Aerospace Model 0118MF would be on science payload interfaces where superior accuracy, calibration stability, and linear resistance are required to ensure data integrity. This comprehensive sensor combination ensures high-fidelity monitoring of every thermal zone while also providing redundancy and survivability in localized sensor degradation and/or failure [10] while ensuring autonomy in TCS by fusing passive stability and active heating.

Radiator sizing radiators and interface management are based on subsystem-level thermal dissipation predictions as expressed by Current Best Estimate (CBE) and Maximum Expected Value (MEV). CBE thermal loads are the nominal thermal loads based on the system specifications. The MEV includes approximately 30% margin to accommodate off-nominal conditions. For example, for the drill, combined CBE is approximately 119.4 W, in multiple drilling modes of auger and percussion, whereas MEV is approximately 155.2 W (~30% thermal margin). The microwave heating subsystem also has the capacity for transient peaks up to 200 W during science runs and factors in the thermal dissipation predictions into the heater redundancy and allocation of the RTG waste heat. These predictions should be used to develop thermal stability under nominal and maximum worst-case thermal predictions under all mission phases. The cold trap is thermally anchored to a radiator specifically sized to remove the heat from the vapor that enters the condenser rapidly without leveraging heavy cryocoolers. Radiator sizing was carried out using Stefan-Boltzmann law for radiative heat transfer, in which ϵ is emissivity, σ is the Stefan-Boltzmann constant ($5.6704 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$), A_{rad} is the radiating area, T is the radiator temperature, T_{sink} is the sink temperature (deep space temperature ~3 K) [17] [18].

$$A_{rad} = \frac{1.3 \times Q_{dissipation}}{\epsilon \sigma A (T^4 - T_{sink}^4)} \quad (1)$$

Given a condenser temperature of 100 K and incorporating a 10 K design margin (resulting in 110 K), the application of the P50 estimate yields a condenser temperature of 105 K; the required radiator area to reject ~0.1 W of heat continuously is calculated to be ~0.0229-0.028 m² for hot and cold cases, respectively. Then the dimensions of the radiator panel are determined to be in the range of 15 cm – 16 cm 16 cm. A 30% growth margin (1.3) was applied to compensate for the uncertainties related to surface contamination, emissivity degradation, and thermal coupling losses. The radiator consists of aluminum (3 mm thick), approximately 0.26 kg in mass. High-emissivity black paints are used on the radiating face of the radiator to increase the efficiency of heat rejection, and low-emissivity coatings are used on other shielded sides to reduce parasitic losses. The radiator-cold trap condenser employs an annealed pyrolytic graphite (APG) material to minimize contact resistance and contains a mechanically actuated heat switch to decouple the radiator from the cold trap during inactive times to avoid uncontrolled excessive overcooling. The method sufficiently releases heat so that sufficient water vapor is captured in the cold trap, operates for further analysis, while not imposing any additional payload mass (cryocooler) or unnecessarily using power.

5.3 Propulsion and Ballistic Hopping

The objective of the propulsion and ballistic hopping subsystem for the lunar hopper vehicle is to facilitate efficient, multi-hop mobility across lunar craters, with a particular emphasis on inter-crater traverses ranging from 0.5 to 6 kilometres per hop.

The gravitational force g is set at 1.62 m/s². The hops are targeted with apex heights (H) ranging from 1500 to 5500 m and horizontal distances (D) spanning 2000 to 6000 m across a total of five hops.

The vehicle baseline has an initial wet mass of 500 kg, a dry mass of 200 kg, and a specific impulse of 317 seconds. The total thrust is 1780 N, which gives a thrust-to-weight ratio of about 2.2 at liftoff on the Moon. Sensors include lidar for finding the terrain and avoiding hazards when descending to the ground, inertial measurement units for guidance, and cameras for optical tracking. The propulsion system includes main engines for hopping and reaction control system (RCS) thrusters for controlling the craft's position in space [Appendix A].

5.3.1 Hopping Architecture

The utilisation of rocket thrusters facilitates propulsive hopping, a technique that has been demonstrated to be particularly efficacious in lunar conditions, enabling the attainment of multi-kilometre ranges with precise targeting through the implementation of thrust vectoring [26]. The technology has been developed over several decades of research and experimentation with vertical takeoff and vertical landing (VTVL), drawing on the heritage of Masten vehicles [27].

The nominal concept of operations (CONOPS) involves an ascent impulse to achieve required vertical and horizontal velocities, followed by an unpowered ballistic coast, and a terminal braking burn to null velocity at landing. This sequence repeats for each hop, with RCS pulses for attitude maintenance and propellant settling pre-burn. Minimum-apex ballistic arcs serve as nominal for efficiency. Lofted ballistic profiles increase apex height for terrain clearance or communications. Hover hops (climb-translate-descend) incur ΔV penalties from gravity losses. Multi-impulse segmented profiles add flexibility but complexity [28].

5.3.2 Ballistic Trajectory Analysis

Ballistic trajectories for lunar hops leverage vacuum conditions and lunar gravity ($g = 1.62 \text{ m/s}^2$). Vertical velocity for apex height $H(i)$ (1500–5500 m) is given by:

$$v_{\text{vert}} = \sqrt{2gH(i)} \quad (2)$$

which is derived from energy conservation given by:

$$1/2mv_{\text{vert}}^2 = mgH(i) \quad (3)$$

The time-of-flight was found to be 86-147 seconds obtained using:

$$t_{f(i)} = 2v_{\text{vert}}/g \quad (4)$$

Horizontal distance $D(i)$ was found to be 2000-6000 m obtained using:

$$v_{\text{horiz}} = D(i)/t_{f(i)} \quad (5)$$

Launch speed is given by:

$$v_0 = \sqrt{v_{\text{vert}}^2 + v_{\text{horiz}}^2} \quad (6)$$

with per-hop ΔV for ascent and descent given by:

$$\Delta v(i) = 2v_0 \quad (7)$$

Trajectories form parabolas given by:

$$x(t) = v_{\text{horiz}} t, z(t) = v_{\text{vert}} t - 0.5(gt^2) \quad (8)$$

Dispersions from Isp ($\pm 1-2\%$), mass ($\pm 5 \text{ kg}$), or cutoff ($\pm 0.1 \text{ s}$) add $\sim 8 \text{ m/s}$ ΔV ; impulsive burns assume short durations ($\sim 20-30 \text{ s}$), ignoring plume-surface interactions [29].

5.3.3 Delta-V Estimation

The hop set targets 0.5-6 km inter-crater traverses. ΔV components include ascent (impart v_0), coast (zero), and braking (null v_0).

Results obtained were:

- Hop 1 ($H=1500 \text{ m}$, $D=2000 \text{ m}$): 104.28 m/s
- Hop 2 (2500 m, 3000 m): 141.45 m/s

- Hop 3 (3500 m, 4000 m): 166.85 m/s
- Hop 4 (4500 m, 5000 m): 187.26 m/s
- Hop 5 (5500 m, 6000 m): 204.85 m/s
- total $\Delta V = 804.69 \text{ m/s}$

T/W evolves from 2.20 (1780 N, 500 kg) to 4.90 (224.31 kg), ensuring liftoff [Appendix 1].

5.3.4 Propellant Budget

Propellant per hop used is given by:

$$m_{\text{prop}}(i) = m_{\text{cur}} - m_{\text{cur}} / \exp(\Delta v(i) / (I_{\text{sp}} g_0)) \quad (9)$$

with $I_{\text{sp}} = 317 \text{ s}$, $g_0 = 9.80665 \text{ m/s}^2$, m_{cur} starting at 500 kg. The value was found to be 41.94–60.83 kg per hop, totaling 275.69 kg. End mass = 224.31 kg, margin = 24.31 kg. RCS (10–20 pulses/hop, 20 N) adds $\sim 14-28 \text{ kg}$; 10% ΔV ($\sim 80 \text{ m/s}$) reserves fit margin. Heat power given by:

$$\dot{Q}_{\text{CBE}} = 0.5 \times \text{thrust}_N \times v_e \times f_{\text{heat}} \quad (10)$$

CBE ($\sim 14.1 \text{ kW}$) gives 1.39–2.01 MJ per hop, totaling 7.50 MJ (CBE) and 13.50 MJ (MEV) [Appendix 1].

5.3.5 Propellant Selection Rationale

MMH/MON-25 was selected as the optimal propellant, driven by mission-specific needs for shadow tolerance and restart reliability. Its storability eliminates the risks of boil-off in prolonged cold ($\leq -30^\circ \text{C}$), in contrast to the use of cryogenics, which incur mass penalties from insulation, refrigeration, and boil-off margins [33, 34, 32]. Hypergolic ignition has been demonstrated to ensure at least 10 restarts without causing wear to auxiliary hardware. This has been proven in Apollo's multi-start engines during periods of extended dormancy [35]. Whilst the lower Isp increases the propellant mass by approximately 30–35% in comparison with LOX/LH₂, the pressure-fed simplicity of the process obviates the necessity for turbopumps and cryocoolers, thereby engendering net mass savings [32]. The considerable experience accumulated over a period of more than five decades in the field of lunar lander development serves to mitigate the risks associated with such endeavours. This is in contrast to the maturation phase of LOX/CH₄ and the absence of lunar lander flights with LOX/LH₂ [30, 31]. The impact of toxicity on ground operations is a significant consideration, and the necessity for SCAPE suits in this regard has been well-documented. However, the potential for autonomous lunar utilisation has been identified as a mitigating factor in these concerns [32].

5.3.6 Thrust band and T/W targets

The lunar hopper has been designed to achieve a total thrust of 1.6–2.0 kN, thereby ensuring a thrust-to-weight (T/W) ratio of ≥ 2 at a 500 kg wet mass, where the lunar weight is approximately 810 N ($500 \times 1.62 \text{ m/s}^2$). This results in a thrust-to-weight ratio of approximately 2.0–2.47, thereby ensuring that short hop burns are employed to minimise delta-V

losses and thermal loads, while concurrently maintaining liftoff capability on the Moon.

5.3.7 Engine Trade off & Selection

Three primary engine configurations were subjected to evaluation: a single 2.0–2.5 kN engine, four 0.5 kN engines, and two 1 kN engines. The single-engine configuration offers simplicity but lacks redundancy ($T/W = 0$ in the event of failure), which poses a risk of mission loss. It is evident that the implementation of four engines is conducive to optimal fault tolerance, as evidenced by the engine-out T/W ratio of approximately 1.85. However, it should be noted that this configuration results in an augmentation of plumbing complexity. The dual 1 kN configuration is notable for its balanced redundancy (engine-out $T/W \approx 1.23$, sufficient for abort or extended burns) and moderate complexity. The configuration that has been selected is two [36] ISE-200 engines (MON-25/MMH, with a mass of approximately 890 N each, an internal combustion ratio of approximately 317 s, and a capacity for more than 1000 restarts). These engines deliver a total of approximately 1.78 kN, a thrust-to-weight ratio of approximately 2.2, and landing-grade throttling [36]. The following constitutes a backup option: It is evident that the Nammo LEROS-4 (900–1100 N, $I_{sp} \geq 321$ s) is less suited for deep throttling [37].

5.3.8 TVC & Throttle Requirements

Each engine requires a gimbal with a range of ± 5 – 10° for the purpose of thrust vector control (TVC), thereby facilitating guidance, navigation, and control (GN&C). Differential throttling is employed to offload RCS demands. A throttle ratio of 10:1 has been shown to ensure precise landing, thereby minimising over-thrust in the final metres (Thruster and RCS selection).

5.3.9 RCS Layout and Sizing

The reaction control system (RCS) adopts an 8-thruster "X" layout (4 pods, 2 thrusters each, canted) using L3Harris ISE-5 (20 N, MON-25/MMH), offering single-fault tolerance and decoupled torques for pitch/yaw/roll with minimal translation coupling. As L3Harris asserts, the utilisation of over 100,000 pulses is instrumental in ensuring longevity. Alternatives to this include ArianeGroup or Moog hydrazine thrusters (1–22 N), which require separate monopropellant tanks [37, 38].

5.4 Communication, Command and Data Handling

The autonomous hopper requires robust, terrain-aware and resilient communication capability. The hopper communicates via short-range wireless link with a commercial lander (Nova-C). This lander is equipped with a Lunar Surface Communication System

(LSCS) providing 4G/LTE connectivity over the lunar surface. Hopper telemetry and command exchange is routed through this link to mission operations on Earth. During link loss, the CD&H system shifts to store-and-forward mode, storing critical data until transmission becomes viable. The router placed at the crater rim helps the hopper inside the crater to communicate with the lander on the moon surface.

S/X/Ka bands can be used to range from 300m to 2km with a fixed terrain-aware tuning antenna. Continuous or burst telemetry can be used to link with Nova C. It can receive commands and have live fault reporting. While inside the crater the hopper enters store mode until the router is visible. The link status is checked pre-hop and post-hop and retry handshake if connection is lost.

Priority Level	Data Types	Mode of Transmission
Critical	Fault logs; EPS anomalies, vapor loss	Immediate transmission; Compressed and timestamped
High	Drilling stats, heating cycles; Sample Summaries	Scheduled bursts
Medium	Spectrometer data, key images;	Bundled; compressed; send back on bandwidth
Low	Raw video; Environmental Telemetry	Stored onboard; Send only if bandwidth allows

Fig. 15. Telemetry Priority Classes

Data Type	Transmission Trigger	How data is send
Health and Fault logs	Immediately on link	Compressed, tagged, sent on high priority
Science Data	Time Triggered	Grouped and continuous uploaded during uplink
Images/Spectra	On Command	Downlink selectively via bandwidth

Fig. 16. Transmission Strategy Summary

The priority in which different types of data is transmitted and the transmission strategy implemented are described in Fig.8. and Fig.9.

The instruments required are On Board Computer (Lander & Hopper), S Band Communication Module (Lander & Router & Hopper), X Band Communication Module (Lander), Cameras (Lander & Hopper). The router employs an S-band communication module (e.g., AAC Clyde Space PULSAR-HSTX) with ~ 5 W nominal power consumption, operating from a 5 V bus, supporting data rates up to 10 Mbps, and weighing ~ 100 g. To ensure correct antenna alignment after spring-based deployment, an orientation mechanism is required, with the complete router system (including compute units, power storage, antenna, insulation, and casing) estimated at 700–1000 g and consuming 10–15 W. The lander carries an onboard computer (e.g., AAC Clyde Space Q7s), consuming ~ 1 W and supporting multiple data interfaces and memory, complemented by S-band and X-band communication modules (e.g., EnduroSat X-band transmitter) capable of up to 150

Mbps datarates. The total C&DH subsystem on the lander is expected to consume ~20 W and weigh 200–500 g. The hopper includes its own onboard computer, S-band communication module, and basic sensors, with estimated mass 200–500 g and power draw of 10–15 W. These estimates exclude external sensors, cameras, drill, and spectrometer subsystems, whose contributions are accounted for separately by the payload team.

5.4.1 Interface between CCDH and other subsystems

Propulsion Subsystem: The CCDH system uploads burn profiles, initiates and terminates thruster firing through pulse count and duration commands, and controls tank operations such as valve opening and closing. Telemetry inputs include tank pressure, temperature, valve status, and real-time estimates of thrust level and Δv . Bulk burn profiles (state vectors and timing tables) are uploaded via file delivery protocols. Communication with the propulsion system uses CAN bus for routine telemetry, with relay-controlled branch circuits supporting 28 V solenoid operation.

Electrical Power Subsystem (EPS): Interfaces allow the CCDH to command power modes, manage solar array tracking, and issue load-shed inhibit signals. It receives telemetry on bus voltages, currents, and battery state-of-charge.

Thermal Control Subsystem: The CCDH sets heater power levels and valve states and monitors thermistor networks as well as pump status.

Payload and Science Subsystems: The CCDH deploys payload elements, commands the start and stop of scanning or heating sequences, and handles large-volume science data streams.

Onboard Data Storage: The CCDH manages memory resources, including formatting, data compression, and controlled data dumps for downlink.

5.5 Electrical Power System

The Electrical Power System is designed around a General Purpose Heat Source Radioisotope Thermoelectric Generator (RTG), providing a continuous baseline output of ~300 W throughout the mission. This steady source of power enables long-duration operations in permanently shadowed regions (PSRs), where solar-based systems are infeasible. The RTG also contributes to thermal resilience by supplying waste heat that can be redistributed across subsystems to reduce heater demand.

To supplement the RTG, a rechargeable battery with a capacity of 500–1000 Wh is included to buffer short-term high-power activities that exceed the RTG's continuous output. Such activities include drilling, microwave heating, and cryotrap operation, which may

draw up to 300–400 W during peak science runs. This hybrid RTG–battery architecture ensures uninterrupted baseline operation while providing flexibility for burst loads without oversizing the primary power source.

Subsystem power requirements were estimated from heritage data and commercial specifications. The hopper's onboard computer (10–15 W), router communication module (~5 W), and lander OBC (~1 W) define the nominal load, while science payloads impose the most demanding conditions. The TRIDENT drill requires ~87 W in auger mode, the microwave heater consumes 150–250 W, the spectrometer requires ~30 W for extended runs, and the cryotrap system is expected to draw 50–200 W depending on cooling duty. Ancillary sensors and cameras each consume only a few watts. These values align with previous system-level allocations and have been cross-validated with laboratory and mission heritage data.

Power flow is managed by a centralized Power Distribution Unit (PDU), which regulates bus voltage, isolates critical loads, and incorporates load-shedding logic for fault tolerance. Redundancy is ensured by dual-string regulation of critical avionics and command/data handling components. The RTG's direct current output is stabilized and routed to both housekeeping loads and battery charging. Battery discharge is scheduled during peak demand, while active balancing prevents deep depletion to ensure cycle longevity.

Mission phases were analyzed to size the EPS appropriately:

- Initialization (post-landing): baseline ~20-30 W, peaks ~117 W (TRIDENT drill test + spectrometer). RTG alone sustains baseline, while the battery covers transients.
- Hopper detachment & router deployment: baseline ~30-40 W, with short actuator bursts. RTG supports baseline; battery provides pulses.
- Science operations: baseline ~25 W, peaks up to ~380 W excluding the cold trap. The RTG sustains continuous loads, while the 500-1000 Wh battery buffers drilling, microwave heating, and spectrometer cycles.
- Data relay and closeout: baseline ~30-40 W, with bursts from communication and potential final hops. RTG supplies baseline, while the battery supports short bursts.

This distribution demonstrates that the RTG can reliably sustain essential functions, while the battery supports transient high-demand tasks, maximizing system efficiency without excessive mass growth. The EPS design thus balances endurance, survivability, and operational flexibility, ensuring that Lunar Vitalis can meet its scientific and exploration objectives within the extreme environment of the lunar south pole.

6. Conclusions

The Lunar Vitalis mission demonstrates a robust approach to exploring permanently shadowed regions of the lunar south pole through a combination of autonomous mobility, subsurface access, and in-situ resource utilization technologies. The mission achieves direct ground-truth measurements of water ice at multiple sites within a single campaign. The operational concept of ballistic hops accessing distinct sampling locations maximizes spatial coverage while validating drilling, extraction, and spectral confirmation methods under true lunar conditions.

Over a surface campaign of 2-4 Earth days, the mission also demonstrates the engineering feasibility of resource acquisition in extreme cryogenic environments and provides critical data for future human and robotic exploration, offering insights into the scalability of water extraction and utilization technologies that will underpin sustainable lunar presence and pave the way for deep space exploration.

Acknowledgements

Our deepest gratitude goes to the SGAC Space Exploration Project Group Co-Leads, whose leadership and support were the backbone of this project.

Appendix A (Hopper Performance Metrics)

<https://github.com/Cosmoshekhar/Lunar-Vitalis-P-BH>

References

- [1] Chakraborty, T., Syed, T. H., Heggy, E., Putrevu, D., & Dutta, U. (2024). On the reachability and genesis of water ice on the Moon. *ISPRS Journal of Photogrammetry and Remote Sensing*, 211, 392-405.
<https://doi.org/10.1016/j.isprsjprs.2024.03.020>
- [2] Zacny, K., Chu, P., Vendiola, V., et al. (2022). TRIDENT Drill for VIPER and PRIME-1 Missions to the Moon. 53rd Lunar and Planetary Science Conference, Abstract 1068.
- [3] Paulsen, G., Mank, Z., Wang, A., et al. (2018). The Regolith and Ice Drill for Exploration of New Terrains (TRIDENT): a One-Meter Drill for the Resource Prospector Mission. *Proceedings of the 44th Aerospace Mechanisms Symposium*, NASA/CP—2018-219887, 13–20.
- [4] Roush, T. L., Colaprete, A., Cook, A., et al. (2021). The VIPER Near Infrared Volatile Spectrometer System (NIRVSS). 52nd Lunar and Planetary Science Conference, Abstract 1678.
- [5] Cook, A. M., Colaprete, A., Roush, T. L., et al. (2016). Multicolor Imagery and NIR Spectroscopy Instrumentation for Planetary Surface Volatile Prospecting. 47th Lunar and Planetary Science Conference, Abstract 2462.
- [6] de Castro, N., & Li, S. (2025). Laboratory experiments on the VNIR spectroscopy of water ice and lunar highland simulant mixtures: Effects of particle size, particle shape, phase angle, and ice abundance. *Icarus*, 435, 116578
- [7] Cole, J. D., Lim, S., Sargeant, H. M., et al. (2023). Water extraction from icy lunar simulants using low power microwave heating. *Acta Astronautica*, 209, 95–103.
- [8] Cole, J. D., Sheridan, S., Lim, S., et al. (2025). The effectiveness of microwave heating as an ISRU extraction technique on different arrangements of icy lunar regolith. *Planetary and Space Science*, 255, 106011.
- [9] Compton, B. M., Krause, T. S., & Struchen Deans, L. M. (2023). An Experimental Study on Low-Pressure Frost Formation for Lunar Polar Water Capture. 52nd International Conference on Environmental Systems (ICES-2023-475), NASA GRC.
- [10] Krause, T. S., Compton, B. M., & Struchen Deans, L. M. (2023). Thermal Model of Ice Growth in Vacuum for Lunar Water Production. NASA GRC Presentation (slides).
- [11] Savoia, M., Rusconi, A., Rizzi, F., et al. (Finmeccanica-Leonardo), Barber, S., Fisackerly, R., Carpenter, J., & Lavagna, M. (ProSEED team).
- [12] Ethridge, E. C. (2014–2015). Microwave Extraction of Volatiles: “Moon in a Bottle” LN₂ Cold Trap Proof-of-Concept and follow-on work. Poster/notes, NASA/MSFC.
- [13] Youhong Liu, Yingzeng Zhang, Qingpei Xiang, Fanhua Hao, Qi An, Haodong Chen, Comprehensive modeling and parametric analysis of Multi-Mission Radioisotope Thermoelectric Generator, *Applied Thermal Engineering*, <https://doi.org/10.1016/j.applthermaleng.2022.119447>
- [14] Zacny, K., Chu, P., Vendiola, V., et al. (2022). TRIDENT Drill for VIPER and PRIME-1 Missions to the Moon. 53rd Lunar and Planetary Science Conference, Abstract 1068.
- [15] L3Harris Technologies, “Bipropellant Rocket Engines — Enabling in-Space Operations and Exploration”, Technical datasheet, May 2025. Available: <https://www.l3harris.com/sites/default/files/2025-05/13harris-ar-bipropellant-rocket-engines.pdf>
- [16] J. P. Williams et al., “Seasonal Polar Temperatures on the Moon,” *JGR Planets*, vol. 124, no. 10, pp. 2505–2521, Oct. 2019, doi: <https://doi.org/10.1029/2019je006028>
- [17] H. Peabody, “Thermal Design for Spaceflight Fall 2022,” ntrs.nasa.gov, Mar. 2023, Available: <https://ntrs.nasa.gov/citations/20230003714>

- [18] G. C. Birur, G. Siebes, T. D. Swanson, and E. I. Powers, "Spacecraft Thermal Control," ntrs.nasa.gov, Jan. 2001, Available: <https://ntrs.nasa.gov/citations/20010091676>
- [19] "VIPER -PROGRAM OF RECORD TECHNICAL INFORMATION TO INFORM PARTNERSHIP PROPOSAL FORMULATION VIPER -Technical Information," 2024. Accessed: Sep. 16, 2025. [Online]. Available: https://assets.science.nasa.gov/content/dam/science/psd/lunar-science/documents/VIPER_Technical_Information_AFPP.pdf
- [20] Spacecraft Thermal Control Handbook, Volume I: Fundamental Technologies | The Aerospace Press, The Aerospace Press, 2025. <https://arc.aiaa.org/doi/book/10.2514/4.989117>
- [21] T.-Y. Park, J.-J. Lee, J.-H. Kim, and H.-U. Oh, "Preliminary Thermal Design and Analysis of Lunar Lander for Night Survival," International Journal of Aerospace Engineering, vol. 2018, pp. 1–13, Oct. 2018, doi: <https://doi.org/10.1155/2018/4236396>
- [22] R. E. Tate, "Light weight radioisotope heater unit (LWRHU): a technical description of the reference design," Los Alamos National Laboratory, LA -9078-MS, 1982.
- [23] "7.0 Thermal Control - NASA." <https://www.nasa.gov/smallsat-institute/sst-soa/thermal-control/#7.3.1>
- [24] P. Fischl, J. Bitanga, and W. Whittaker, "THERMAL MODELING AND DESIGN OF A MICRO-ROVER FOR LUNAR POLAR EXPLORATION." Accessed: Sep. 16, 2025. [Online]. Available: <https://www.hou.usra.edu/meetings/isairas2020fullpapers/pdf/5058.pdf>
- [25] "ISL73590SEH - Radiation Hardened, 2-Terminal Temperature Transducer," Renesas, 2025. <https://www.renesas.com/en/products/isl73590seh> (accessed Sep. 16, 2025)
- [26] "NASA Jet Propulsion Laboratory," NASA, Accessed: Sep. 09, 2025. [Online]. Available: <https://www-robotics.jpl.nasa.gov>
- [27] "CORE," CORE, Accessed: Sep. 09, 2025. [Online]. Available: <https://core.ac.uk>
- [28] "DSpace @ MIT," MIT Libraries, Accessed: Sep. 09, 2025. [Online]. Available: <https://dspace.mit.edu>
- [29] "NASA Technical Reports Server," NASA, Accessed: Sep. 09, 2025. [Online]. Available: <https://ntrs.nasa.gov>
- [30] B. Thombare, S. Shekhar, and V. Malhotra, "On the effect of energetic material induction in cryogenic propellants under elevated pressure conditions," in AIP Conference Proceedings, vol. 2341, p. 030014, 2021, doi: 10.1063/5.0049976.
- [31] P. Mhatre, B. Thombare, S. Shekhar, and P. S., "ENERGETIC HYDROCARBON BASED ROCKET PROPELLANTS FOR UPPER STAGE PROPULSION," ResearchGate, 2021.
- [32] P. Madsen, "Project SPECTRA: Experimental Evaluation of a Liquid Storable Propellant," Copenhagen Suborbitals, Rev. 1.0, Sep. 11, 2012
- [33] "Next-Generation Moon Thrusters Withstand Over 60 Hot-Fire Tests," SciTechDaily, Accessed: Sep. 09, 2025. [Online]. Available: <https://scitechdaily.com/next-generation-moon-thrusters-withstand-over-60-hot-fire-tests/>
- [34] H. P. Trinh, C. Burnside, and H. Williams, "Assessment of MON-25/MMH Propellant System for Deep-Space Engines," NASA Marshall Space Flight Center, Huntsville, AL, Tech. Rep. 20190033304, 2019
- [35] P. Gajjar and V. Malhotra, "Advanced upper stage energetic propellants," in 2018 IEEE Aerospace Conference, Big Sky, MT, USA, 2018, pp. 1–12, doi: 10.1109/AERO.2018.8396709
- [36] "L3Harris," L3Harris Technologies, Inc., Accessed: Sep. 09, 2025. [Online]. Available: <https://www.l3harris.com>
- [37] "Moog monoprop family," Moog Inc., Accessed: Sep. 09, 2025. [Online]. Available: <https://www.moog.com>
- [38] "space-propulsion.com," ArianeGroup, Accessed: Sep. 09, 2025. [Online]. Available: <https://www.space-propulsion.com>